

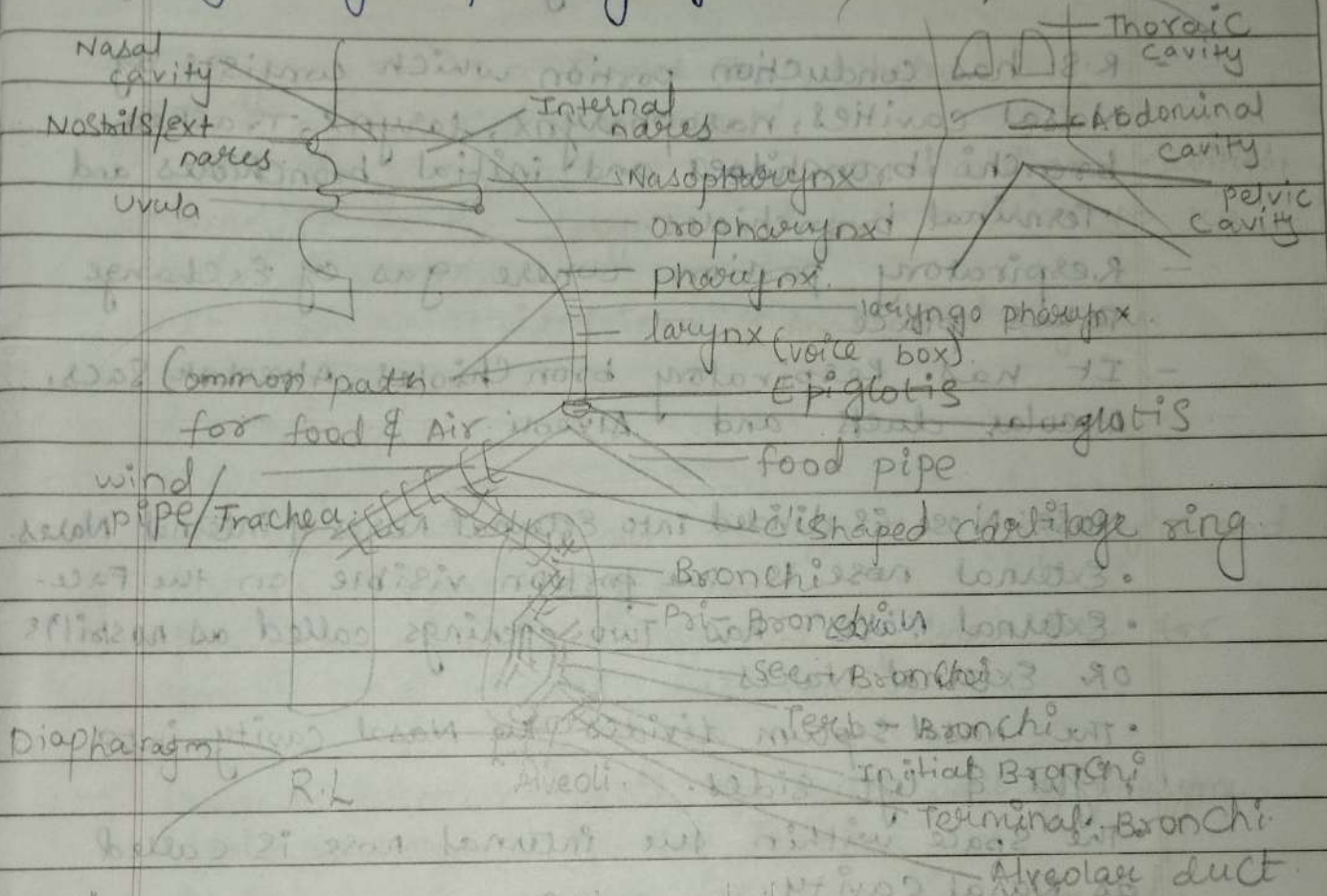
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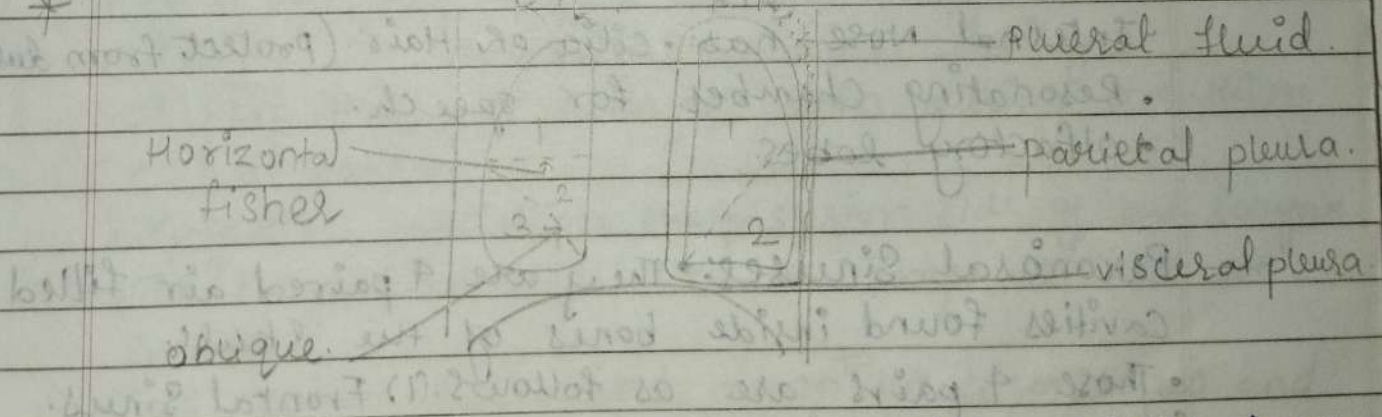
* RESPIRATORY SYSTEM:-

• Organs of respiratory system

division of cavities in Entire H.B



*



- A Human Respiratory System consist of Nose, pharynx, larynx, Trachea, lungs, Bronchus, Bronchioles, and Alveoli.
- Majorly respiratory system we divide as upper respiratory system and lower respiratory system.

- From which upper respiratory system consists of.
 - Nose, pharynx, larynx and Trachea.
- And lower r.s. consist of Bronchi and lungs.
- R.S. has conduction portion which consists of Nasal cavities, nasopharynx, larynx, Trachea, bronchi, bronchioles and initial bronchioles and Terminal bronchioles.
- Respiratory portion where gas of Exchange takes place.
- It has Respiratory bronchioles, Alveolar Sacs, Alveolar ducts and Alveoli.
- Nose: Nose is divided into External Nares & internal Nares.
 - External nose is the portion visible on the Face.
 - External Nose has Two openings called as nostrils OR External Nares.
 - The nasal septum divides the Nasal cavity into right & left sides.
 - The space within the internal nose is called as Nasal cavity.
 - Internal nose has:
 - cilia OR Hair. (protect from dust)
 - Resonating Chamber for speech.
 - olfactory lobes.
- Paranasal Sinuses:
 - They are 4 paired air filled cavities found inside bones of the skull.
 - Those 4 pairs are as follows. (1) Frontal Sinus.
 - (2) Ethmoidal Sinus.
 - (3) Sphenoidal Sinus.
 - (4) Maxillary Sinus.
 - Mucosal line the paranasal Sinuses

- The Sinuses reduce the weight of the skull and serve as a resonant chamber to affect the quality of the voice.

- Pharynx: It is a funnel shaped tube.

- It is also called as Throat.

- It lies just posterior to the nasal and oral cavities, superior to the larynx and just anterior to the cervical vertebra.

- It can be divided into 3 Anatomical regions:

1. Nasopharynx

2. Oropharynx

3. Laryngopharynx.

- Both food and air pass through the oropharynx and through laryngopharynx below.

- pharynx provides a resonating chamber for speech and houses the tonsils.

- Larynx: Larynx connects the lower part of pharynx.

- laryngopharynx to the Trachea.

- It is the key organ for producing sound.

- Larynx is comprised of nine cartilages.

- It keeps the air passages open during breathing and digestion.

- Epiglottis is located on the posterior side of the larynx.

- The Epiglottis closes like a Trap door as we swallow.

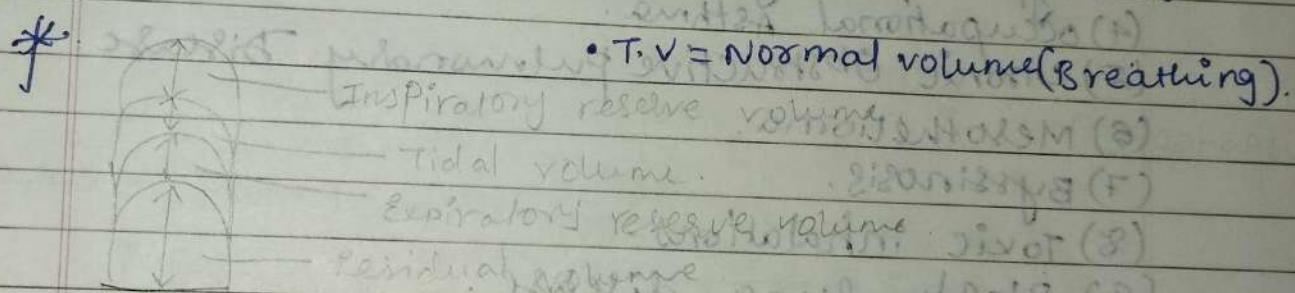
- This action steers food down to the Esophagus and away from the windpipe.

- Inside the larynx there is vocal cords which have Elastic Ligaments at their core.

- **Trachea:** Also called as windpipe.
 - Tubular passage way for air. (12cm long & 2.5 cm diameter).
 - The inner wall of Trachea is lined with ciliated mucous membrane with many goblet cells.
 - which serves to trap the incoming particles.
 - It is located anterior to the Esophagus.
 - The 16-20 'C' shaped solid cartilage rings provides semirigid support so that the tracheal wall does not collapse inward.
 - The Trachea divides into right and left primary bronchi an internal ridge called the carina.
- **Bronchi:** Trachea divides into right primary bronchus which goes to R.L. and left primary bronchus which goes to L.L.
 - after entering the lungs primary bronchi divides into secondary bronchi then tertiary and then initial bronchioles and terminal bronchioles.
- **Lungs:** Lungs are paired cone-shaped organs in the thoracic cavity.
 - Right lung has 3 lobes and it is slightly larger than left lung because it has 2 lobes.
 - Each lung is protected by double layered serous membrane called as pleural membrane.
 - Superficial layer is called as parietal pleura.
 - wall of thoracic cavity is called as visceral pleura. Covers the lungs themselves.

- Alveoli: There are more than 300 million alveoli in both the lungs.
 - They called as microscopic air sacs.
 - oxygen passes through the thin membrane of the alveoli and into the blood stream.
 - RBC pick up the oxygen and carry it to the body's organs and tissues.

- In 1 minute human breathe 12-17 Times.
- T.V $\rightarrow$ Tidal volume. (6000 ml - 8000 ml).
- I.R.V $\rightarrow$ Inspiratory reserve volume (2500 ml to 3000 ml).
- E.R.V $\rightarrow$ Expiratory reserve volume (1000 ml - 1100 ml).
- R.V $\rightarrow$ Residual volume. (1100 ml - 1200 ml).



- E.R.V always present in lungs.
- R.V also always present in lungs but, can't remove it normally. Even after death.
- Except if body crushes down then only R.V gets released.

- I.C $\rightarrow$ Inspiratory capacity: $T.V + I.R.V = 3000 - 3500 \text{ ml}$
- E.C $\rightarrow$ Expiratory Capacity: $T.V + E.R.V = 1500 \text{ to } 1600 \text{ ml}$
- F.R.C $\rightarrow$ Functional residual capacity: $E.R.V + R.V = 2100 \text{ to } 2300 \text{ ml}$.
- V.C $\rightarrow$ vital capacity: $E.R.V + T.V + I.R.V$

$$\begin{aligned} & \text{or } E.R.V + I.C \\ & \text{or } E.C + I.R.V \end{aligned} = 4000 - 4600 \text{ ml}$$
- Total lungs: $T.V + T.R.V + F.R.V + R.V = 5100 - 5800 \text{ ml}$

- Respiratory Disorders: 1) Asthma:- A condition in which a person's airways become inflamed, narrow and swell and produce extra mucus, which makes it difficult to breathe.

2) Emphysema: A disorder affecting the alveolar walls (Alveoli [Tiny air sacs]) of the lungs.

3) Pneumonia: Infection that inflames air sacs in one or both lungs which may fill with fluid.

4) Occupational R.D: 1) Silicosis (2) Asbestosis

(3) Hypersensitivity pneumonitis.

(4) Occupational Asthma.

(5) Chronic Obstructive Pulmonary Disease.

(6) Mesothelioma.

(7) Byssinosis.

(8) Toxic inhalations.

(9) Black lung disease.

(10) Lung Cancer.

(11) Bronchiolitis.

(12) Parenchymal disease.

(13) Reactive airway disease.

(14) Lung Cancer.

• To prevent diseases, person should wear mask and should have to be careful while working.